

Study G — Appendix A: Initial Conditions and Vehicle Geometry

Every state flown, and the engine layout it was flown with

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1 Which initial conditions belong to which fault

All of them, and they are the same ones. The campaign is a *paired* design: one set of initial conditions is drawn per regime and then reused, unchanged, for every one of the sixteen plant configurations. That is deliberate and

it is what makes the study’s comparisons work — because the state distribution is literally identical across faults, any difference in landing rate is attributable to the plant alone, and the healthy-vs-fault comparison becomes a within-sample McNemar test on discordant pairs rather than a contrast between two independently drawn samples. So there is no per-fault initial-condition list to give. There is one list, of $5 \times 10 = 50$ states, reproduced in full in section 4; section 5 then shows what each fault did from each of them.

2 The vehicle and where its engines are

The lander carries **two independently gimballed TVC engines**, each rated at half the nominal DPS thrust, mounted symmetrically either side of the centreline in a plane below the centre of gravity. Body axes are x forward, y right, z **down**, so a positive z engine station is *below* the CG.

Engine	Body station (x, y, z) [m]	Thrust [N]	Hover share [N]	Gimbal [°]	Role
Engine 1 (index 0)	(+0.00, -1.50, +2.50)	2280 – 22520	6265	±6.0	healthy in every case
Engine 2 (index 1)	(+0.00, +1.50, +2.50)	2280 – 22520	6265	±6.0	carries the fault

Every fault in the catalogue acts on **engine 2** (index 1, body station y = +1.50 m). Engine 1 stays healthy throughout, so each case is an *asymmetry* the vehicle has to trim as well as a loss of performance. The vehicle is laterally symmetric, so which engine is chosen does not matter — only that it is one of two.

The lateral half-spacing $y_{eng} = 1.50$ m is the parameter behind the engine-out result: a single gimbal can trim a one-engine-out roll asymmetry only while $y_{eng} \leq dz_{eng} \tan \delta_{max} = 0.263$ m, and this vehicle is built at 1.50 m — a factor of 5.7 outside it.

Vehicle parameter	Value
Mass	7711 kg
Inertia I_{xx}, I_{yy}, I_{zz}	5368, 5368, 5040 kg · m ²
Lunar gravity	1.625 m/s ²
Hover thrust (total)	12530 N
DPS envelope (total)	4560 – 45040 N
Engine plane below CG, dz_{eng}	2.50 m
Engine half-spacing, y_{eng}	1.50 m
Thrust lag τ_T / gimbal ω_n, ζ	0.40 s / 4.0 rad/s, 0.70
RCS	4 quads x 4 thrusters, 445 N each, arm 1.70 m
Glide-slope cone	30° from horizontal
Contact altitude (engine cut)	1.0 m

3 The sampling box of each regime

Initial conditions are scrambled Sobol points in these boxes, rejected if they fall outside the approach cone or are already past the hard loss-of-control criteria. **upset** additionally keeps only draws that are genuinely upset (a body rate of at least 10 °/s on some axis, or at least 15° of tilt).

3.1 approach — On approach

high, descending, small dispersions — the vehicle is where a healthy descent would have put it, and the fault is the only thing wrong.

State	Unit	lower	upper
x_E	m	-350	+350
y_E	m	-350	+350
alt	m	+800	+1300
u	m/s	-20	+0
v	m/s	-5	+5
w	m/s	+0	+25
phi	deg	-8	+8
theta	deg	-8	+8
psi	deg	-15	+15
p	deg/s	-3	+3
q	deg/s	-3	+3
r	deg/s	-3	+3
T1	N	+5325	+7205
dp1	deg	-1.5	+1.5
dy1	deg	-1.5	+1.5
dp1_dot	deg/s	-3	+3
dy1_dot	deg/s	-3	+3
T2	N	+5325	+7205
dp2	deg	-1.5	+1.5
dy2	deg	-1.5	+1.5
dp2_dot	deg/s	-3	+3
dy2_dot	deg/s	-3	+3

3.2 dispersed — Dispersed

Study F's wide box: position, velocity, attitude, rates and both sets of actuator states drawn over their full admissible ranges.

State	Unit	lower	upper
x_E	m	-900	+900
y_E	m	-900	+900
alt	m	+300	+1400
u	m/s	-25	+10
v	m/s	-12	+12
w	m/s	-5	+28
phi	deg	-25	+25
theta	deg	-25	+25
psi	deg	-40	+40
p	deg/s	-25	+25
q	deg/s	-25	+25
r	deg/s	-25	+25
T1	N	+2280	+2.252e+04
dp1	deg	-6	+6
dy1	deg	-6	+6
dp1_dot	deg/s	-14.4	+14.4
dy1_dot	deg/s	-14.4	+14.4
T2	N	+2280	+2.252e+04
dp2	deg	-6	+6
dy2	deg	-6	+6
dp2_dot	deg/s	-14.4	+14.4
dy2_dot	deg/s	-14.4	+14.4

3.3 upset — Upset

the corner of the box where the vehicle is already rotating or tilted (≥ 10 deg/s on some axis, or ≥ 15 deg of tilt) with the gimbals deflected — the states a fault transient actually produces.

State	Unit	lower	upper
x_E	m	-600	+600
y_E	m	-600	+600
alt	m	+500	+1100
u	m/s	-30	+15
v	m/s	-15	+15
w	m/s	+0	+30
phi	deg	-40	+40
theta	deg	-40	+40
psi	deg	-60	+60
p	deg/s	-30	+30
q	deg/s	-30	+30
r	deg/s	-30	+30
T1	N	+2280	+2.252e+04
dp1	deg	-6	+6
dy1	deg	-6	+6
dp1_dot	deg/s	-14.4	+14.4
dy1_dot	deg/s	-14.4	+14.4
T2	N	+2280	+2.252e+04
dp2	deg	-6	+6
dy2	deg	-6	+6
dp2_dot	deg/s	-14.4	+14.4
dy2_dot	deg/s	-14.4	+14.4

3.4 low_late — Low and late

between 60 m and 200 m and still descending: the fault arrives when there is very little altitude left to trade for time.

State	Unit	lower	upper
x_E	m	-200	+200
y_E	m	-200	+200
alt	m	+60	+200
u	m/s	-25	+10
v	m/s	-14	+14
w	m/s	+5	+32
phi	deg	-28	+28
theta	deg	-28	+28
psi	deg	-40	+40
p	deg/s	-16	+16
q	deg/s	-16	+16
r	deg/s	-16	+16
T1	N	+3759	+9398
dp1	deg	-4	+4
dy1	deg	-4	+4
dp1_dot	deg/s	-8	+8
dy1_dot	deg/s	-8	+8
T2	N	+3759	+9398
dp2	deg	-4	+4
dy2	deg	-4	+4

State	Unit	lower	upper
dp2_dot	deg/s	-8	+8
dy2_dot	deg/s	-8	+8

3.5 critical — Critical

below 140 m, descending hard, tilted and rotating — the regime where even the healthy vehicle frequently has no trajectory left, so the fault has to be paid for out of a margin that is already spent.

State	Unit	lower	upper
x_E	m	-140	+140
y_E	m	-140	+140
alt	m	+50	+140
u	m/s	-30	+15
v	m/s	-18	+18
w	m/s	+8	+34
phi	deg	-35	+35
theta	deg	-35	+35
psi	deg	-50	+50
p	deg/s	-22	+22
q	deg/s	-22	+22
r	deg/s	-22	+22
T1	N	+3133	+1.002e+04
dp1	deg	-6	+6
dy1	deg	-6	+6
dp1_dot	deg/s	-12	+12
dy1_dot	deg/s	-12	+12
T2	N	+3133	+1.002e+04
dp2	deg	-6	+6
dy2	deg	-6	+6
dp2_dot	deg/s	-12	+12
dy2_dot	deg/s	-12	+12

4 The initial conditions

Positions are in the landing-pad frame (x north, y east) with **altitude positive up**; velocities are body-frame; angles are degrees. T1/T2 are the two engines' thrust states and **dp/dy** their pitch and yaw gimbal deflections, with **_dot** the corresponding gimbal rates. Sampling the actuator states directly is the point of the 22-dimensional box: a fault leaves the gimbals deflected and the thrust away from trim, and those are states no 12-dimensional sampling can reach.

4.1 approach — On approach (10 states)

4.1.1 Rigid-body states

#	x_E [m]	y_E [m]	alt [m]	u [m/s]	v [m/s]	w [m/s]	phi [deg]	theta [deg]	psi [deg]	p [deg/s]	q [deg/s]	r [deg/s]
0	-107	58	1108	-8.15	2.86	12.86	0.58	-0.42	-10.46	0.71	-2.11	-0.31
1	333	-47	862	-13.72	-2.73	3.67	-0.64	7.11	2.99	-0.95	1.12	1.85
2	108	196	1249	-2.58	-1.43	9.21	6.56	2.18	8.35	-2.10	1.88	0.97
3	-331	-186	1003	-18.06	1.58	24.26	-6.74	-5.37	-0.93	2.61	-0.89	-2.50
4	-261	336	869	-16.07	0.59	15.74	5.70	0.06	12.31	-0.68	2.81	-2.23
5	3	-346	1115	-0.82	-0.16	0.69	-5.51	-7.50	-4.89	0.99	-0.32	0.67

#	x_E [m]	y_E [m]	alt [m]	u [m/s]	v [m/s]	w [m/s]	phi [deg]	theta [deg]	psi [deg]	p [deg/s]	q [deg/s]	r [deg/s]
6	260	110	978	-10.40	-3.85	12.19	3.66	-2.29	-14.00	2.14	-2.67	2.89
7	-5	-121	1224	-5.22	4.31	21.38	-3.60	5.22	6.53	-2.58	0.18	-1.33
8	-70	239	927	-1.51	-3.35	21.92	4.16	1.75	13.13	2.98	0.52	-0.51
9	195	-227	1180	-16.61	3.48	6.86	-4.10	-6.69	-5.72	-1.73	-2.34	2.02

4.1.2 Actuator states

#	T1 [N]	dp1 [deg]	dy1 [deg]	dp1_dot [deg/s]	dy1_dot [deg/s]	T2 [N]	dp2 [deg]	dy2 [deg]	dp2_dot [deg/s]	dy2_dot [deg/s]
0	7126	0.94	-1.48	0.00	2.16	6964	1.31	1.02	-0.89	-0.96
1	6214	-0.24	0.72	-0.23	-2.62	5554	-0.44	-0.67	1.00	1.99
2	6551	0.56	1.26	2.58	-1.00	6303	-1.30	0.00	-2.39	0.44
3	5639	-1.26	-0.48	-2.44	0.59	5833	0.43	-1.31	2.50	-2.61
4	6414	-0.72	-0.11	-2.17	1.02	6048	-0.94	1.21	-0.29	-0.30
5	5505	1.42	0.88	2.03	-0.58	6518	0.31	-0.11	0.08	2.65
6	6791	-0.78	0.35	-1.16	-2.19	5740	0.95	0.56	-1.79	1.29
7	5882	0.08	-1.10	1.39	2.58	7150	-0.32	-1.12	1.58	-1.75
8	5414	-0.96	1.35	-1.77	-1.20	5916	-0.84	0.23	-0.66	2.25
9	6267	0.26	-0.57	1.53	0.02	6386	0.16	-1.46	0.45	-0.70

4.2 dispersed — Dispersed (10 states)

4.2.1 Rigid-body states

#	x_E [m]	y_E [m]	alt [m]	u [m/s]	v [m/s]	w [m/s]	phi [deg]	theta [deg]	psi [deg]	p [deg/s]	q [deg/s]	r [deg/s]
0	-275	150	978	-4.26	6.86	11.98	1.83	-1.31	-27.90	5.91	-17.59	-2.62
1	278	503	1287	5.49	-3.44	7.16	20.50	6.81	22.26	-17.53	15.63	8.04
2	-852	-478	746	-21.61	3.79	27.02	-21.07	-16.77	-2.49	21.77	-7.38	-20.81
3	8	-890	993	8.57	-0.37	-4.09	-17.22	-23.43	-13.05	8.22	-2.69	5.59
4	668	282	692	-8.20	-9.24	11.09	11.45	-7.15	-37.33	17.81	-22.28	24.10
5	-12	-311	1234	0.86	10.34	23.23	-11.23	16.33	17.42	-21.47	1.53	-11.09
6	-181	615	579	7.36	-8.04	23.93	13.00	5.47	35.01	24.81	4.33	-4.28
7	502	-584	1136	-19.06	8.35	4.06	-12.80	-20.90	-15.25	-14.41	-19.51	16.86
8	178	40	543	-2.57	5.29	2.94	6.87	-11.02	-34.52	-11.12	-5.52	9.71
9	-506	-13	1103	-11.35	-4.94	15.07	-6.28	15.51	14.60	2.78	20.70	-22.28

4.2.2 Actuator states

#	T1 [N]	dp1 [deg]	dy1 [deg]	dp1_dot [deg/s]	dy1_dot [deg/s]	T2 [N]	dp2 [deg]	dy2 [deg]	dp2_dot [deg/s]	dy2_dot [deg/s]
0	21666	3.77	-5.91	0.01	10.38	19921	5.23	4.09	-4.25	-4.63
1	15475	2.25	5.04	12.37	-4.80	12805	-5.18	0.01	-11.46	2.13
2	5661	-5.06	-1.91	-11.69	2.82	7746	1.72	-5.25	12.01	-12.51
3	4210	5.68	3.53	9.73	-2.80	15125	1.22	-0.42	0.37	12.72
4	18061	-3.13	1.38	-5.56	-10.52	6745	3.80	2.26	-8.59	6.19
5	8268	0.31	-4.42	6.69	12.40	21925	-1.27	-4.50	7.57	-8.42
6	3234	-3.84	5.39	-8.49	-5.75	8641	-3.37	0.90	-3.17	10.82
7	12417	1.02	-2.28	7.36	0.07	13700	0.65	-5.86	2.16	-3.37
8	9853	-2.17	-4.80	-4.34	7.87	3106	3.32	3.20	-10.38	-10.34

#	T1 [N]	dp1 [deg]	dy1 [deg]	dp1_dot [deg/s]	dy1_dot [deg/s]	T2 [N]	dp2 [deg]	dy2 [deg]	dp2_dot [deg/s]	dy2_dot [deg/s]
9	19031	4.98	1.77	5.01	-13.31	18285	-0.60	-2.06	9.35	4.30

4.3 upset — Upset (10 states)

4.3.1 Rigid-body states

#	x_E [m]	y_E [m]	alt [m]	u [m/s]	v [m/s]	w [m/s]	phi [deg]	theta [deg]	psi [deg]	p [deg/s]	q [deg/s]	r [deg/s]
0	-183	100	870	-3.34	8.57	15.43	2.92	-2.09	-41.85	7.09	-21.11	-3.15
1	571	-81	574	-15.87	-8.18	4.41	-3.21	35.53	11.97	-9.50	11.21	18.48
2	185	336	1038	9.20	-4.30	11.05	32.80	10.90	33.38	-21.03	18.76	9.65
3	-568	-319	743	-25.64	4.73	29.11	-33.71	-26.83	-3.73	26.12	-8.86	-24.97
4	-448	576	583	-21.16	1.78	18.89	28.51	0.30	49.24	-6.76	28.14	-22.32
5	6	-593	878	13.16	-0.47	0.82	-27.55	-37.49	-19.57	9.87	-3.23	6.70
6	445	188	714	-8.40	-11.55	14.63	18.32	-11.44	-55.99	21.37	-26.74	28.92
7	-8	-208	1009	3.25	12.93	25.66	-17.98	26.12	26.13	-25.76	1.83	-13.31
8	-121	410	652	11.61	-10.05	26.30	20.80	8.75	52.51	29.78	5.19	-5.14
9	335	-389	956	-22.36	10.44	8.24	-20.48	-33.44	-22.87	-17.29	-23.41	20.23

4.3.2 Actuator states

#	T1 [N]	dp1 [deg]	dy1 [deg]	dp1_dot [deg/s]	dy1_dot [deg/s]	T2 [N]	dp2 [deg]	dy2 [deg]	dp2_dot [deg/s]	dy2_dot [deg/s]
0	21666	3.77	-5.91	0.01	10.38	19921	5.23	4.09	-4.25	-4.63
1	11853	-0.96	2.89	-1.13	-12.60	4742	-1.76	-2.67	4.81	9.56
2	15475	2.25	5.04	12.37	-4.80	12805	-5.18	0.01	-11.46	2.13
3	5661	-5.06	-1.91	-11.69	2.82	7746	1.72	-5.25	12.01	-12.51
4	14005	-2.87	-0.42	-10.42	4.88	10066	-3.76	4.84	-1.39	-1.44
5	4210	5.68	3.53	9.73	-2.80	15125	1.22	-0.42	0.37	12.72
6	18061	-3.13	1.38	-5.56	-10.52	6745	3.80	2.26	-8.59	6.19
7	8268	0.31	-4.42	6.69	12.40	21925	-1.27	-4.50	7.57	-8.42
8	3234	-3.84	5.39	-8.49	-5.75	8641	-3.37	0.90	-3.17	10.82
9	12417	1.02	-2.28	7.36	0.07	13700	0.65	-5.86	2.16	-3.37

4.4 low_late — Low and late (10 states)

4.4.1 Rigid-body states

#	x_E [m]	y_E [m]	alt [m]	u [m/s]	v [m/s]	w [m/s]	phi [deg]	theta [deg]	psi [deg]	p [deg/s]	q [deg/s]	r [deg/s]
0	-61	33	146	-4.26	8.00	18.89	2.05	-1.46	-27.90	3.78	-11.26	-1.68
1	62	112	186	5.49	-4.01	14.95	22.96	7.63	22.26	-11.22	10.01	5.15
2	2	-198	148	8.57	-0.44	5.74	-19.28	-26.24	-13.05	5.26	-1.72	3.58
3	148	63	110	-8.20	-10.78	18.17	12.82	-8.01	-37.33	11.40	-14.26	15.42
4	-3	-69	179	0.86	12.07	28.10	-12.58	18.29	17.42	-13.74	0.98	-7.10
5	-40	137	96	7.36	-9.38	28.67	14.56	6.13	35.01	15.88	2.77	-2.74
6	112	-130	166	-19.06	9.74	12.41	-14.34	-23.41	-15.25	-9.22	-12.49	10.79
7	40	9	91	-2.57	6.17	11.49	7.70	-12.34	-34.52	-7.12	-3.53	6.21
8	-112	-3	162	-11.35	-5.77	21.42	-7.04	17.37	14.60	1.78	13.25	-14.26
9	-152	89	199	-14.97	-2.19	25.56	3.71	-5.74	-20.10	-15.70	-6.53	-8.96

4.4.2 Actuator states

#	T1 [N]	dp1 [deg]	dy1 [deg]	dp1_dot [deg/s]	dy1_dot [deg/s]	T2 [N]	dp2 [deg]	dy2 [deg]	dp2_dot [deg/s]	dy2_dot [deg/s]
0	9160	2.51	-3.94	0.01	5.77	8674	3.49	2.72	-2.36	-2.57
1	7435	1.50	3.36	6.87	-2.67	6691	-3.46	0.01	-6.37	1.18
2	4297	3.79	2.35	5.41	-1.55	7338	0.82	-0.28	0.21	7.07
3	8156	-2.09	0.92	-3.09	-5.85	5003	2.54	1.51	-4.77	3.44
4	5427	0.21	-2.94	3.71	6.89	9232	-0.85	-3.00	4.20	-4.68
5	4025	-2.56	3.60	-4.72	-3.19	5531	-2.25	0.60	-1.76	6.01
6	6583	0.68	-1.52	4.09	0.04	6941	0.44	-3.91	1.20	-1.87
7	5869	-1.45	-3.20	-2.41	4.37	3989	2.21	2.13	-5.77	-5.74
8	8426	3.32	1.18	2.78	-7.39	8218	-0.40	-1.37	5.20	2.39
9	6170	1.89	0.18	1.50	-4.27	8983	3.76	1.10	-3.36	4.36

4.5 critical — Critical (10 states)

4.5.1 Rigid-body states

#	x_E [m]	y_E [m]	alt [m]	u [m/s]	v [m/s]	w [m/s]	phi [deg]	theta [deg]	psi [deg]	p [deg/s]	q [deg/s]	r [deg/s]
0	-43	23	105	-3.34	10.28	21.38	2.56	-1.83	-34.87	5.20	-15.48	-2.31
1	43	78	131	9.20	-5.16	17.58	28.70	9.53	27.82	-15.42	13.76	7.08
2	1	-138	107	13.16	-0.56	8.71	-24.10	-32.80	-16.31	7.24	-2.37	4.92
3	104	44	82	-8.40	-13.86	20.68	16.03	-10.01	-46.66	15.67	-19.61	21.21
4	-2	-48	126	3.25	15.51	30.24	-15.73	22.86	21.77	-18.89	1.34	-9.76
5	-28	96	73	11.61	-12.07	30.79	18.20	7.66	43.76	21.84	3.81	-3.77
6	78	-91	118	-22.36	12.53	15.14	-17.92	-29.26	-19.06	-12.68	-17.17	14.83
7	28	6	70	-1.16	7.93	14.25	9.62	-15.43	-43.16	-9.79	-4.85	8.54
8	-79	-2	116	-12.45	-7.41	23.82	-8.79	21.72	18.25	2.44	18.22	-19.61
9	-106	62	139	-17.11	-2.81	27.80	4.64	-7.18	-25.12	-21.59	-8.98	-12.32

4.5.2 Actuator states

#	T1 [N]	dp1 [deg]	dy1 [deg]	dp1_dot [deg/s]	dy1_dot [deg/s]	T2 [N]	dp2 [deg]	dy2 [deg]	dp2_dot [deg/s]	dy2_dot [deg/s]
0	9733	3.77	-5.91	0.01	8.65	9139	5.23	4.09	-3.55	-3.86
1	7626	2.25	5.04	10.31	-4.00	6716	-5.18	0.01	-9.55	1.77
2	3790	5.68	3.53	8.11	-2.33	7506	1.22	-0.42	0.31	10.60
3	8506	-3.13	1.38	-4.64	-8.77	4653	3.80	2.26	-7.16	5.16
4	5172	0.31	-4.42	5.57	10.33	9822	-1.27	-4.50	6.31	-7.01
5	3457	-3.84	5.39	-7.08	-4.79	5298	-3.37	0.90	-2.64	9.01
6	6584	1.02	-2.28	6.13	0.06	7021	0.65	-5.86	1.80	-2.81
7	5711	-2.17	-4.80	-3.62	6.56	3414	3.32	3.20	-8.65	-8.61
8	8836	4.98	1.77	4.18	-11.09	8582	-0.60	-2.06	7.79	3.58
9	6079	2.84	0.28	2.25	-6.40	9517	5.64	1.65	-5.04	6.54

5 What each fault did from each of them

One grid per regime: rows are the sixteen plants, columns are the numbered initial conditions above. The columns are the *same states* in every row, which is what makes a row-to-row difference a statement about the plant.

L landed · g flew but missed the touchdown gate · . no trajectory found · x already lost before the planner ran

5.1 approach — On approach

Fault	0	1	2	3	4	5	6	7	8	9	landed
Healthy (control)	L	L	L	L	L	L	L	L	L	L	10/10
Thrust reduction, $\eta=0.50$	L	L	L	L	L	L	L	L	L	L	10/10
Thrust reduction, $\eta=0.15$	.	g	.	.	.	g	.	.	.	.	0/10
Thrust excess, $\eta=1.30$	L	L	L	L	L	L	L	L	L	L	10/10
Valve stuck open (thrust floor)	L	L	L	L	L	L	L	L	L	L	10/10
Engine out	.	.	.	.	.	.	.	.	.	.	0/10
Slow thrust response, $\tau_T=2.5$ s	L	L	L	L	L	L	L	L	L	L	10/10
Gimbal bandwidth loss, $\omega_n=0.6$	L	L	L	L	L	L	L	L	L	L	10/10
Gimbal underdamped, $\zeta=0.25$	L	L	L	L	L	L	L	L	L	L	10/10
Gimbal seizure	L	L	L	L	L	L	L	L	L	L	10/10
TVC effectiveness loss (35 %)	L	L	L	L	L	L	L	L	L	L	10/10
Thrust-vector misalignment (3° , 2°)	L	L	L	L	L	L	L	L	L	L	10/10
Thrust oscillation (chugging)	L	L	L	L	L	L	L	L	L	L	10/10
Transport delay (1 interval)	L	L	L	L	L	L	L	L	L	L	10/10
Mixture-ratio shift (coupled)	L	L	L	L	L	L	L	L	L	L	10/10
Throat-erosion drift	L	L	L	L	L	L	L	L	L	L	10/10

5.2 dispersed — Dispersed

Fault	0	1	2	3	4	5	6	7	8	9	landed
Healthy (control)	L	L	L	L	L	L	L	L	L	L	10/10
Thrust reduction, $\eta=0.50$	L	L	g	L	L	L	L	L	L	L	9/10
Thrust reduction, $\eta=0.15$	.	g	.	g	.	.	.	g	g	.	0/10
Thrust excess, $\eta=1.30$	L	L	L	L	L	L	L	L	L	L	10/10
Valve stuck open (thrust floor)	L	L	L	g	L	L	L	g	L	L	8/10
Engine out	.	.	.	.	.	.	.	.	.	.	0/10
Slow thrust response, $\tau_T=2.5$ s	L	L	L	L	L	L	L	L	L	L	10/10
Gimbal bandwidth loss, $\omega_n=0.6$	L	.	.	.	L	.	.	.	L	L	4/10
Gimbal underdamped, $\zeta=0.25$	L	.	.	L	L	L	L	L	L	L	8/10
Gimbal seizure	L	L	g	L	L	L	L	L	L	L	9/10
TVC effectiveness loss (35 %)	L	L	L	L	L	L	L	L	L	L	10/10
Thrust-vector misalignment (3° , 2°)	L	L	L	L	L	L	L	L	L	L	10/10
Thrust oscillation (chugging)	L	L	L	L	L	L	L	L	L	L	10/10
Transport delay (1 interval)	L	L	L	L	L	.	L	L	L	L	9/10
Mixture-ratio shift (coupled)	L	L	g	L	L	L	L	L	L	L	9/10
Throat-erosion drift	L	L	g	g	L	L	L	L	L	L	8/10

5.3 upset — Upset

Fault	0	1	2	3	4	5	6	7	8	9	landed
Healthy (control)	L	L	L	L	L	L	L	L	L	L	10/10
Thrust reduction, $\eta=0.50$	L	L	L	g	g	L	L	L	L	g	7/10
Thrust reduction, $\eta=0.15$	.	.	.	.	.	.	.	.	.	g	0/10
Thrust excess, $\eta=1.30$	L	L	L	L	L	L	L	L	L	L	10/10
Valve stuck open (thrust floor)	L	L	L	g	L	L	L	L	L	g	8/10
Engine out	.	.	.	.	.	.	.	.	.	.	0/10
Slow thrust response, $\tau_T=2.5$ s	L	L	L	g	L	L	L	.	L	L	8/10
Gimbal bandwidth loss, $\omega_n=0.6$	L	L	.	.	L	.	L	.	.	.	4/10
Gimbal underdamped, $\zeta=0.25$	L	L	.	.	L	L	L	L	L	L	8/10

Fault	0	1	2	3	4	5	6	7	8	9	landed
Gimbal seizure	L	L	L	g	L	L	L	L	L	L	9/10
TVC effectiveness loss (35 %)	L	L	L	L	L	L	L	L	L	L	10/10
Thrust-vector misalignment (3°, 2°)	L	L	L	L	L	L	L	L	L	L	10/10
Thrust oscillation (chugging)	L	L	L	L	L	L	L	L	L	L	10/10
Transport delay (1 interval)	L	L	L	L	L	L	L	.	L	L	9/10
Mixture-ratio shift (coupled)	L	L	L	g	L	L	L	L	L	L	9/10
Throat-erosion drift	L	L	L	g	L	L	L	L	L	g	8/10

5.4 low_late — Low and late

Fault	0	1	2	3	4	5	6	7	8	9	landed
Healthy (control)	L	L	L	L	L	.	L	L	L	.	8/10
Thrust reduction, $\eta=0.50$	.	L	L	L	.	.	L	L	.	.	5/10
Thrust reduction, $\eta=0.15$	.	.	.	.	.	.	.	.	.	.	0/10
Thrust excess, $\eta=1.30$	L	L	L	L	L	.	L	L	L	.	8/10
Valve stuck open (thrust floor)	L	L	L	L	L	.	L	L	L	.	8/10
Engine out	.	.	.	.	.	.	.	.	.	.	0/10
Slow thrust response, $\tau_T=2.5$ s	L	L	L	L	L	.	L	L	.	.	7/10
Gimbal bandwidth loss, $\omega_n=0.6$	L	.	L	L	L	.	L	L	L	.	7/10
Gimbal underdamped, $\zeta=0.25$	L	L	L	L	L	.	L	L	L	.	8/10
Gimbal seizure	L	L	L	L	L	.	L	L	L	.	8/10
TVC effectiveness loss (35 %)	L	L	L	L	L	.	L	L	L	.	8/10
Thrust-vector misalignment (3°, 2°)	L	L	L	L	L	.	L	L	L	.	8/10
Thrust oscillation (chugging)	L	L	L	L	L	.	L	L	L	.	8/10
Transport delay (1 interval)	L	L	L	L	L	.	L	L	.	.	7/10
Mixture-ratio shift (coupled)	L	L	L	L	L	.	L	L	.	.	7/10
Throat-erosion drift	L	L	L	L	L	.	L	L	L	.	8/10

5.5 critical — Critical

Fault	0	1	2	3	4	5	6	7	8	9	landed
Healthy (control)	L	L	.	L	L	.	L	L	.	.	6/10
Thrust reduction, $\eta=0.50$	.	L	.	.	.	.	L	.	.	.	2/10
Thrust reduction, $\eta=0.15$	.	.	.	.	.	.	.	.	.	.	0/10
Thrust excess, $\eta=1.30$	L	L	.	L	L	.	L	L	.	.	6/10
Valve stuck open (thrust floor)	L	L	.	L	L	.	L	L	.	.	6/10
Engine out	.	.	.	.	.	.	.	.	.	.	0/10
Slow thrust response, $\tau_T=2.5$ s	.	L	.	L	.	.	L	.	.	.	3/10
Gimbal bandwidth loss, $\omega_n=0.6$	L	.	.	L	.	.	.	L	.	.	3/10
Gimbal underdamped, $\zeta=0.25$	L	.	.	L	L	.	L	L	.	.	5/10
Gimbal seizure	.	L	.	L	L	.	L	L	.	.	5/10
TVC effectiveness loss (35 %)	L	L	.	L	L	.	L	L	.	.	6/10
Thrust-vector misalignment (3°, 2°)	L	L	.	L	L	.	L	L	.	.	6/10
Thrust oscillation (chugging)	L	L	.	L	L	.	L	L	.	.	6/10
Transport delay (1 interval)	.	L	.	L	.	.	L	.	.	.	3/10
Mixture-ratio shift (coupled)	.	L	.	L	.	.	L	.	.	.	3/10
Throat-erosion drift	L	L	.	L	L	.	L	L	.	.	6/10

6 Files

results/G_initial_conditions.csv the 50 states above, one row each

results/G_states.npz
results/G_samples.csv

the same states plus every regime's box
one row per solve (state x plant)